

EESC2020 - Climate System Dynamics

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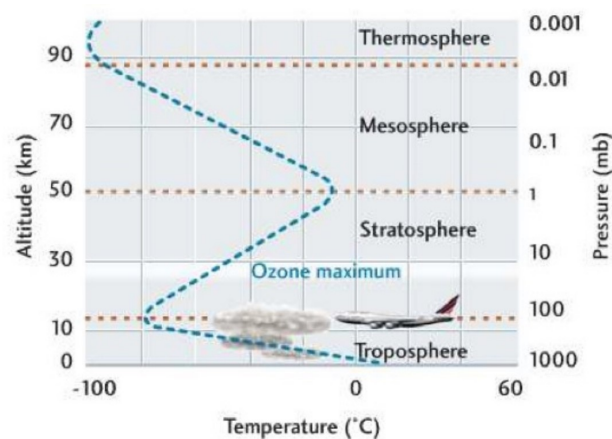
Lecture 1 - Introduction

Climate and Weather

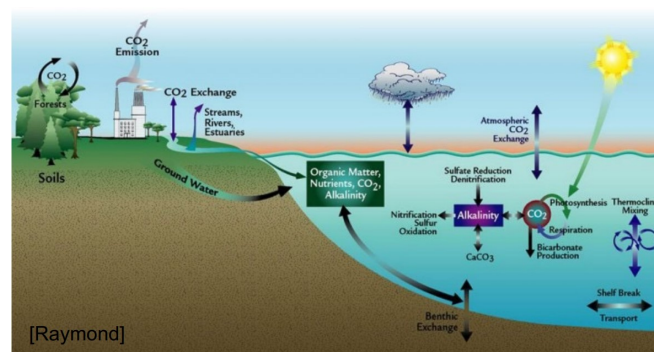
- Weather: the transient state of the atmosphere over short periods (from hours to weeks).
- Climate: the statistical description in terms of the mean and variability of **relevant quantities** over a period of time (months to thousands or millions of years). WMO defines an averaging period of 30 years (IPCC, 2021).
 - Temperature
 - Humidity
 - Precipitation
 - Cloud
 - Pressure
 - Wind

Climate system

- Atmosphere
 - Thermosphere
 - Mesosphere
 - Stratosphere
 - Troposphere



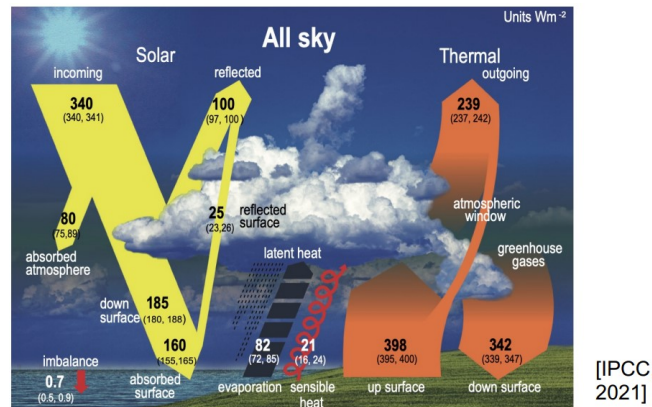
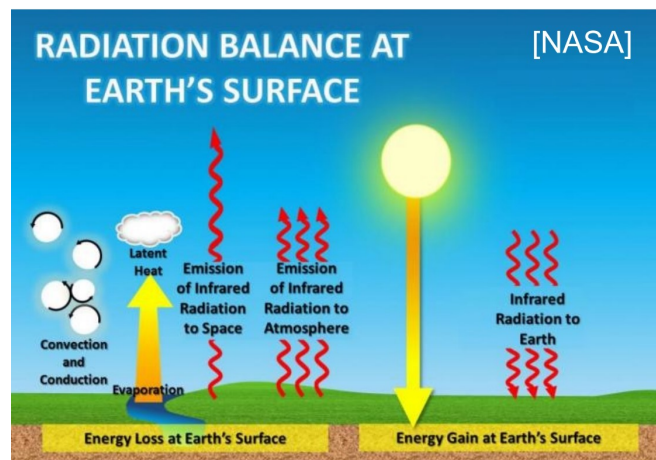
- Hydrosphere
 - Evaporation
 - Transpiration
 - Condensation
 - Precipitation
 - Snowmelt runoff
 - Surface runoff
- Cryosphere
 - Glaciers
 - Ice sheets
 - Snow cover
 - Sea ice
- Lithosphere
 - Plate tectonics
 - Rock cycle
 - $\Rightarrow$ Geological carbon cycle
- Biosphere
 - Biogeophysical effects: Energy balance and exchange of momentum
 - Biogeochemical effects: absorption of trace gases



Lecture 2 - Energy Balance

Components of Energy Balance in Climate Systems

- Solar radiation
- Atmosphere absorption
- Evaporation
- Surface (Absorption/Reflection)
- Greenhouse gases
- Evaporation & Latent heat
- Convection and conduction



Parameters Related to Energy and Radiations

Wave length-speed equation: $c = \lambda \nu$

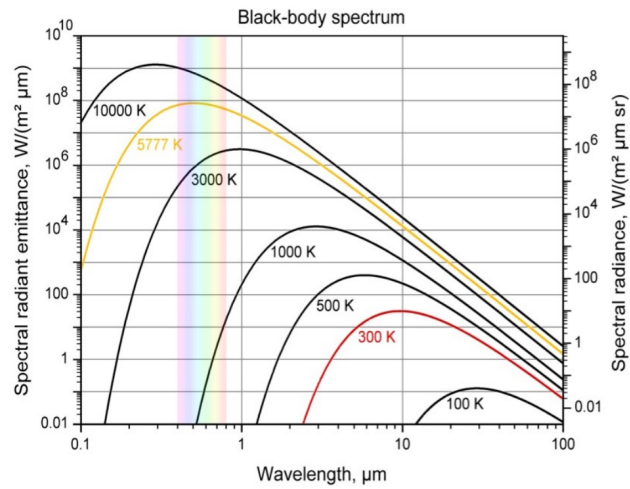
Photon energy equation: $E = \hbar \nu$, $\hbar = 6.626 \times 10^{-34} \text{ kg m}^2 \text{ s}^{-1}$

Total radiant flux density: energy per unit time per unit area per unit wavelength emitted by a blackbody

Planck's function:

$$F(\lambda, T) = \frac{2\pi\hbar c^2}{\lambda^5 [\exp(\frac{\hbar c}{\lambda k T}) - 1]}, k = 1.381 \times 10^{-23} \text{ J K}^{-1} \text{ (Boltzmann constant)}$$

- λ_{max} moves left as T increases, along with higher radiative flux
- 300 K corresponds to the temperature of the Earth, with $\lambda_{max} \approx 10 \mu\text{m}$ (thermal IR)
- 5777 K corresponds to the temperature of the Sun, with $\lambda_{max} \approx 0.5 \mu\text{m}$ (visible light)



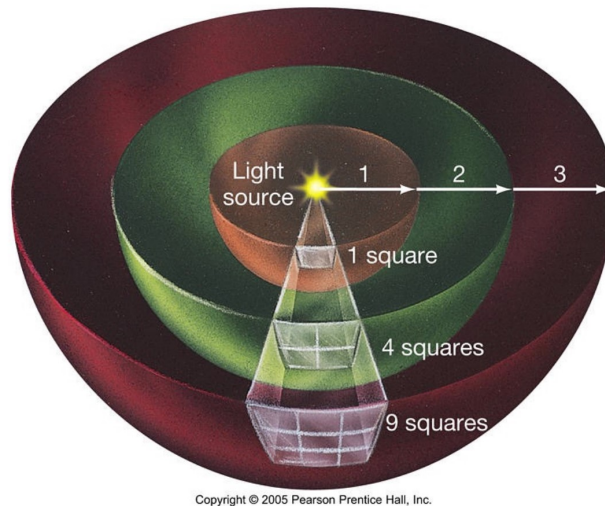
$$\frac{\partial F}{\partial \lambda} = 0 \Rightarrow \lambda_{max} = \frac{(2.898 \times 10^{-3} \text{ m K})}{T}$$

Stefan-Boltzmann Law:

$$F(T) = \sigma T^4, \text{ W} \cdot \text{m}^{-2} = \text{J} \cdot \text{s}^{-1} \cdot \text{m}^{-2}$$

Luminosity: total power emitted by a star.

$$L = (4\pi R_S^2)(\sigma T^4) = 3.87 \times 10^{26} \text{ W}$$



Simple Climate Model

- Energy balance: shortwave (solar) and longwave (terrestrial) radiation
- Albedo: reflection of shortwave radiation by clouds, ice and the land surface
- Greenhouse effect: absorption and reemission of longwave radiation by clouds and gases in the atmosphere

Incoming Solar Flux

- Solar constant $S_0 = 1361 \text{ W m}^{-2}$
- Average incoming solar flux per unit surface area = $\frac{S_0}{4}$

Albedo

- Fraction of incident solar radiation of all wavelengths reflected by a body or surface

Energy Balance of the Earth

- Input (shortwave): $E_{abs} = \frac{S_0}{4}(1 - \alpha_p)$
- Output (longwave): $E_{emit} = \sigma T_e^4$

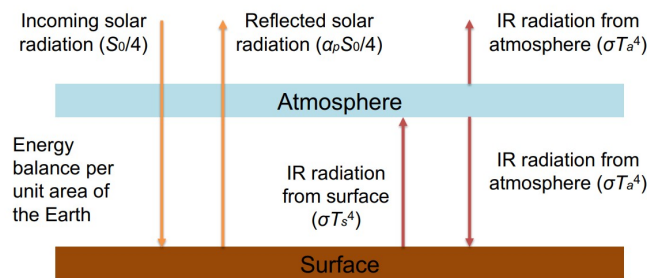
– Let

$$E_{abs} = E_{emit} \Rightarrow \frac{S_0}{4}(1 - \alpha_p) = \sigma T_e^4 \Rightarrow T_e = \left[\frac{S_0(1 - \alpha_p)}{4\sigma} \right]^{1/4} \text{ (effective temperature of a planet)}$$

- Effective planetary temperature $T_e \approx 255 \text{ K}$
- Global mean surface temperature $T_s \approx 288 \text{ K}$

- Greenhouse effect absorbs and re-emits terrestrial (infrared) radiation, $T_s > T_e$
- Molecules absorbing IR Radiation

- * Water vapor
- * CO₂
- * N₂O
- * CH₄
- * O₃



Energy Balance of the Atmosphere

- Input: $E_{abs} = \sigma T^4$ (from the surface)
- Output: $E_{emit} = 2\sigma T_a^4$ (to outer space and back to surface)

–

$$T_s = 2^{1/4} T_a$$

(Atmosphere is always colder than the surface)

Energy Balance at the Surface

- Input: $E_{abs} = (S_0/4)(1 - \alpha_p) + \sigma T_a^4$ (from the sun and from the atmosphere)
- Output: $E_{emit} = \sigma T_s^4$ (to the atmosphere)

—

$$T_a = T_e$$

—

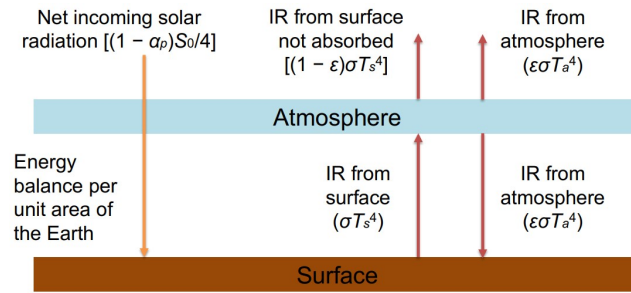
$$T_s = 2^{1/4} T_e$$

Leaky Atmosphere

- Atmosphere does allow part of the IR radiation to pass through into space
- Convection transports energy from surface to atmosphere

Lecture 3 - Climate Feedback

Leaky Atmosphere



- Input: $E_{abs} = (S_0/4)(1 - \alpha_p) + \epsilon\sigma T_a^4$
- Output: $E_{emit} = \sigma T_s^4$

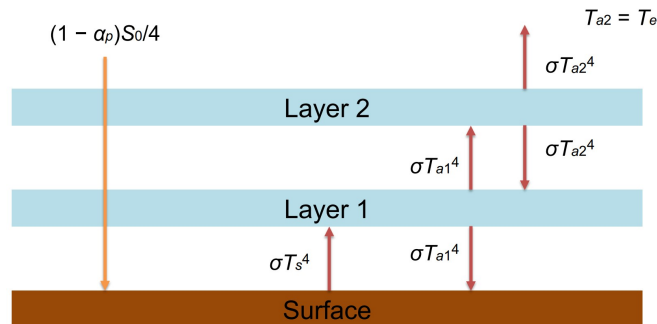
—

$$T_a = \left(\frac{2}{2 - \epsilon} \right)^{1/4} T_e$$

—

$$T_s = \left(\frac{2}{2 - \epsilon} \right)^{1/4} T_e = \left[\frac{S_0}{4\sigma} (1 - \alpha_p) \left(\frac{2}{2 - \epsilon} \right) \right]^{1/4}$$

Multilayer Model



•

$$T_s = (N + 1)^{1/4} T_e, N = \frac{\epsilon}{2 - \epsilon}$$

Radiative Forcing and Climate Sensitivity

Radiative Forcing ΔQ : net downward radiative flux.

Climate Sensitivity ΔT_s : change in temperature due to external driver of climate change, $\Delta T_s = \lambda \Delta Q$.

$$[\Delta T_s]_{\text{final}} = \frac{[\Delta T_s]_{\text{initial}}}{1 - f} = \frac{\lambda_0 \Delta Q}{1 - f}$$

Climate Feedback Mechanisms

- Water vapor feedback: $T \uparrow \rightarrow \text{evaporation} \uparrow \rightarrow \text{atmospheric water vapor} \uparrow \rightarrow \text{longwave absorption} \uparrow \rightarrow T \uparrow$

- Cloud feedback:

$T \uparrow \rightarrow \text{evaporation} \uparrow \rightarrow \text{atmospheric water vapor} \uparrow \rightarrow \text{cloudiness} \uparrow \rightarrow \text{albedo} \uparrow \rightarrow T \uparrow$

$T \uparrow \rightarrow \text{evaporation} \uparrow \rightarrow \text{atmospheric water vapor} \uparrow \rightarrow \text{cloudiness} \uparrow \rightarrow \text{longwave absorption} \uparrow \rightarrow T \uparrow$

Low-altitude clouds tend to affect albedo and shortwave (-) more, but high-altitude clouds affect longwave (+) more.

- Ice-albedo feedback

$T \uparrow \rightarrow \text{sea ice and glaciers} \uparrow \rightarrow \text{albedo} \downarrow \rightarrow T \uparrow$

- Biogeochemical feedback:

$T \uparrow \rightarrow \text{photosynthesis} \uparrow \rightarrow \text{CO}_2 \downarrow \rightarrow T \downarrow$

$T \uparrow \rightarrow \text{respiration} \uparrow \rightarrow \text{CO}_2 \uparrow \rightarrow T \uparrow$

$T \uparrow \rightarrow \text{photosynthesis} \downarrow \rightarrow \text{CO}_2 \uparrow \rightarrow T \uparrow$

- Biogeophysical feedback:

$T \uparrow \rightarrow \text{photosynthesis} \uparrow \rightarrow \text{albedo} \downarrow \rightarrow T \uparrow$

$T \uparrow \rightarrow \text{photosynthesis} \downarrow \rightarrow \text{albedo} \uparrow \rightarrow T \downarrow$

$T \uparrow \rightarrow \text{photosynthesis} \uparrow \rightarrow \text{evapotranspiration} \uparrow \rightarrow T \downarrow$

$T \uparrow \rightarrow \text{photosynthesis} \downarrow \rightarrow \text{evapotranspiration} \downarrow \rightarrow T \uparrow$

Stability of Climate

Stable: a small perturbation will be dampened and the system returns to its original state.

Unstable: a small perturbation will be amplified and the system will diverge to a new state (either to “explode” or to reach a new equilibrium).

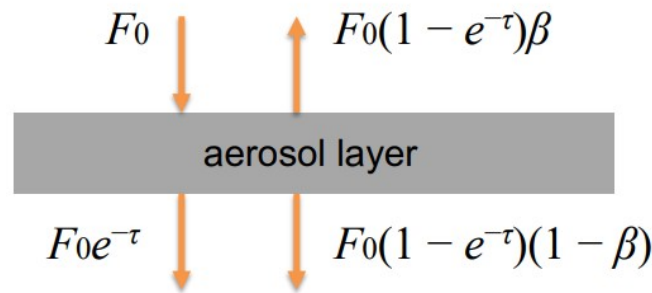
Lecture 5 - Climate Variability

Ice-albedo feedback

- Case study: Venus (Surface temperature $\approx 490^\circ\text{C}$)
 - Venus's atmosphere consists mostly of CO_2 , $\bar{M} = 44 \text{ g/mol}$, it has much thicker atmosphere than the Earth's, the surface pressure is 90 times of the Earth's.
- **Runaway greenhouse effect:** climate instability associated with water vapor feedback might have caused Venus to evolve from an Earth-like state to today's hellish state.
- Case study: Snowball Earth
 - Earth has experienced several "ice house" and "hot house" periods throughout its history since 4.5 billion years ago.
 - Earth has hypothetically undergone periods of being completely or nearly frozen at least once - known as **Snowball Earth**.
 - * Triggered by an unusually cold period, ice-albedo feedback rendered climate unstable: more ice cover $\rightarrow$ more solar reflection $\rightarrow$ cooler surface $\rightarrow$ more ice cover
 - * Triggered by some initial melting (orbital change) $\rightarrow$ more solar absorption $\rightarrow$ warmer surface $\rightarrow$ more melting
 - * It is believed that the rise of O_2 generating bacteria has triggered the initial cooling or warming (biochemistry perspective).
- Ice ages: Not the whole Earth surface was frozen; It was believed that the Sun causes the changes instead of CO_2 , we see temperature decline preceded CO_2 .
 - Ice-albedo feedback played important role for abrupt warming from glacial to interglacial periods.
 - Note that sea ice melting does not change sea level, only land ice sheet melting does.
 - Ice-albedo feedback could amplify and worsen warming in the Arctic.

Aerosols and Climate

Mountain Pinatudo Eruption in Philippines in 1991 injected nearly 20 million tons of SO_2 , and was oxidized in the stratosphere (10-50 km above ground) and formed sulfate (SO_4^{2-}) particles that scattered solar radiation.



F_0 = incident solar radiation

τ = aerosol optical depth

$e^{-\tau}$ = fraction of F_0 transmitted

$(1 - e^{-\tau})$ = fraction of F_0 scattered

β = fraction of scattered radiation that is scattered upward

Aerosol albedo:

$$\alpha_a = \beta(1 - e^{-\tau})$$

Increased sulfur emissions from China during the 2000s partially offset the rising CO_2 forcing. Oxidized SO_2 will form ice-state H_2SO_4 which can enhance albedo.

Problems of Aerosols to mitigate Climate Change:

- Delivery and lifespan issues (PM 2.5)
- Disruption of hydrological cycle
- Stratospheric ozone depletion
- Disruption of ecosystems
- Appearance of the sky

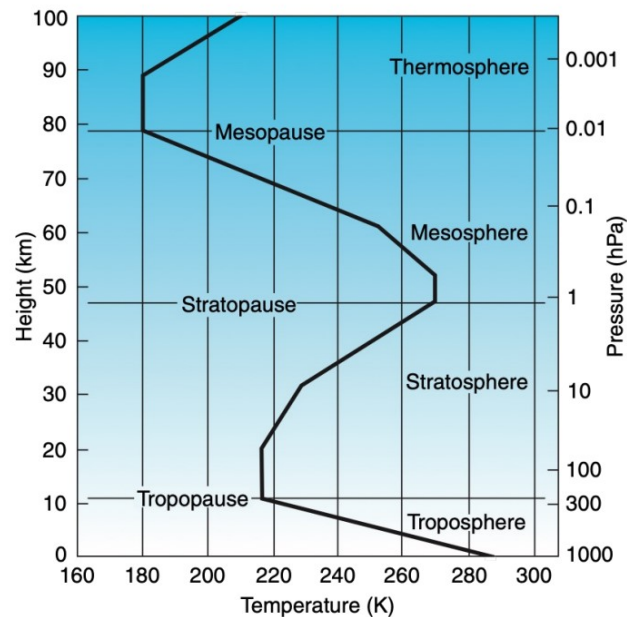
Geoengineering: (1) solar radiation management; (2) CO_2 removal

Seasonality of the Earth's Climate

Seasons occur since the Earth is tilted, and cause variations in duration of solar insolation, angle of solar insolation. Meridional imbalance of radiant energy cause energy transpotation from low to high latitudes to maintain the energy equilibrium. The transport can be done by atmospheric and ocean circulation.

Lecture 6 - Hydrostatic Balance

Vertical Structure of the Atmosphere



- Thermosphere: absorption of high-energy UV in association with dissociation and ionization of O_2 and N_2 molecules; too few greenhouse gases to have effective IR loss.
 - Ionosphere (upper part of the thermosphere) is important for reflection of radio waves
- Mesosphere: where the temperature meets maximum at the stratopause due to heavy UV absorption by ozone (O_3).
- Stratosphere: stable, stratified layer where the UV-absorbing ozone layer exists (Chapman cycle):
 - $O_2 + h\nu \rightarrow 2O$ (slow cycle)
 - $O + O_2 + M \rightarrow O_3 + M$ (fast cycle)
 - $O_3 + h\nu \rightarrow O + O_2$ (fast cycle)
 - $O + O_3 \rightarrow 2O_2$ (slow cycle)
- Troposphere: 85% of total atmospheric mass and nearly all of its water, lapse rate = $|\frac{dT}{dz}| \approx 6.5^\circ C \text{ km}^{-1}$ caused by radiative transfer (between layers of atmosphere) and mechanical transfer (convection).

Vertical Pressure Profile for Atmosphere and Ocean

- Atmosphere: pressure decreases exponentially with increasing altitude,

$$p = p_0 \exp\left(-\frac{z - z_0}{H}\right)$$

- Ocean: Pressure increases linearly with depth,

$$p = p_0 + \rho gh$$

- Hydrostatic Balance Equation:

$$\frac{dp}{dz} = -\rho g$$

- Ideal Gas Law:

$$pV = nR^*T \Rightarrow p = \rho RT, R = R^*/M_d,$$

where M_d represents the molar mass of air

- Scale Height:

$$H = \frac{RT}{g}$$

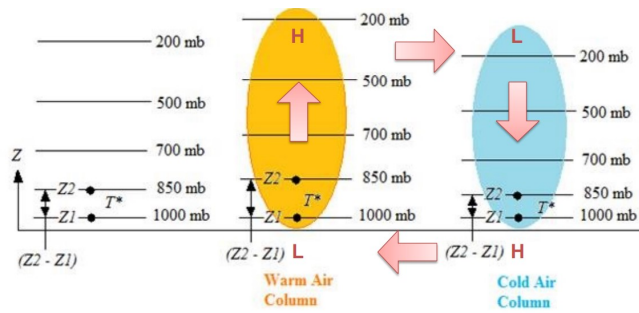
- Barometric Law:

$$p = p_0 \exp\left(-\frac{z - z_0}{H}\right)$$

- Hypsometric Equation:

$$\Delta z = \frac{R\bar{T}}{g} \ln\left(\frac{p_1}{p_2}\right)$$

- Differential heating can induce a horizontal pressure gradient driving horizontal motion and circulation in the vertical plane.



Lecture 7 - Convection

- Buoyancy of a fluid parcel:

$$b = g \frac{\rho_{env} - \rho}{\rho}$$

- Buoyancy of an air parcel:

$$b = g \frac{T - T_{env}}{T_{env}}$$

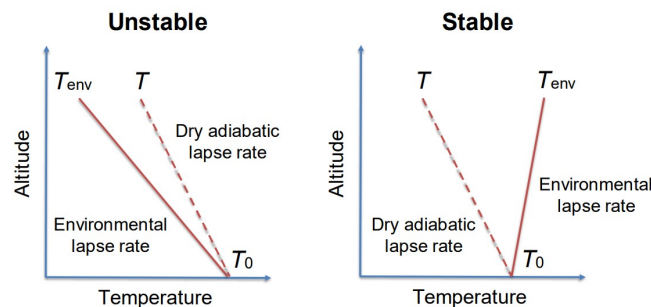
- Internal energy of an ideal gas is only a function of its temperature
- $dU = mc_V dT$ if a gas is confined to a constant volume, and all heat absorbed are used to raise its T .
- First Law of Thermodynamics: change in internal energy of a system must be equal to the sum of 1) heat applied to the system; 2) work done on the system. $dU = \delta Q - pdV$. If the system expands, $dU < \delta Q$; if the system contracts: $dU > \delta Q$.
- For a parcel of an ideal gas and assuming that it undergoes adiabatic motion that ($\delta Q = 0$), then

$$\frac{T_2}{T_1} = \left(\frac{p_2}{p_1} \right)^{\frac{R}{c_p}}, \quad \frac{V_2}{V_1} = \left(\frac{p_1}{p_2} \right)^{\frac{c_V}{c_p}},$$

with hydrostatic balance equation and ideal gas law, we get

$$\frac{dT}{dz} = -\frac{g}{c_p} = -\Gamma_d.$$

- Stability of a Dry Atmosphere



A compressible atmosphere is unstable if its temperature falls with height faster than the dry adiabatic lapse rate.

- Potential Temperature (θ) of an air parcel at pressure p and temperature T is the temperature this parcel would acquire if brought adiabatically to a reference pressure p_0 ,

$$\theta = T \left(\frac{p_0}{p} \right)^{\frac{R}{c_p}}$$

- Troposphere lapse rate is generally stable to dry convection.
- Stability and air pollution dispersion
 - During daytime, solar heating renders unstable atmosphere, convections will occur.
 - During nighttime, surface cooling and low winds help create an inversion layer and trap air pollutants near the surface.

Lecture 8 - Cloud Formation

Clausius-Clapeyron equation

$$\frac{de_s}{dT} = \frac{Le_s}{R_v T^2},$$

approximation: $e_s = 6.108e^{0.0668 \cdot T}$, valid for $0 < T < 25^\circ\text{C}$

Relative Humidity

$$\text{RH} = \frac{e}{e_s(T)} \times 100\%$$

Specific Humidity

$$q = \frac{\rho_v}{\rho} \approx 0.622 \frac{e}{p}$$

Dew Point Temperature

$$e_s(T_d) = e$$

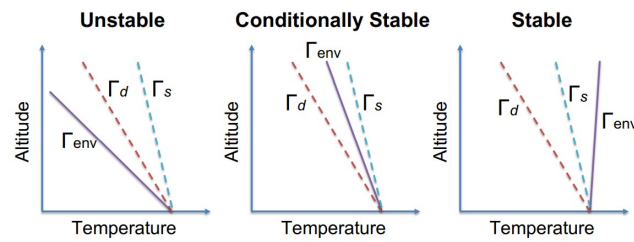
Moist Air Convection

- An air parcel is saturated if
 - Its water partial pressure e equals saturation vapor pressure e_s
 - Its RH reaches 100
 - Its temperature T equals its dew point T_d

It then condenses to liquid water droplets, forming clouds and fogs.

- When an initially unsaturated air parcel rises from surface:
 - q is conserved
 - e drops with height because p also drops with height
 - T_d , function of e would also drop with height. Typical dew point temperature lapse would be $-2^\circ\text{C}\cdot\text{km}^{-1}$
 - Its T drops according to dry adiabatic lapse rate
- T of a parcel always falls with height faster than its T_d , this will make the parcel saturated, forming the cloud base. Then, after reaching saturation, T falls less fast than dry adiabatic lapse rate because of condensation of water releasing latent heat (less internal energy required to rise and expand).
- Temperature of saturated air falls with height at the moist adiabatic lapse rate (Γ_s), which is always smaller than the dry adiabatic lapse rate (Γ_d) for unsaturated air.
- Cloud tops at a certain height when
 - T of saturated air parcel equals that of surrounding
 - All water vapor in the parcel has condensed

- Atmospheric Stability

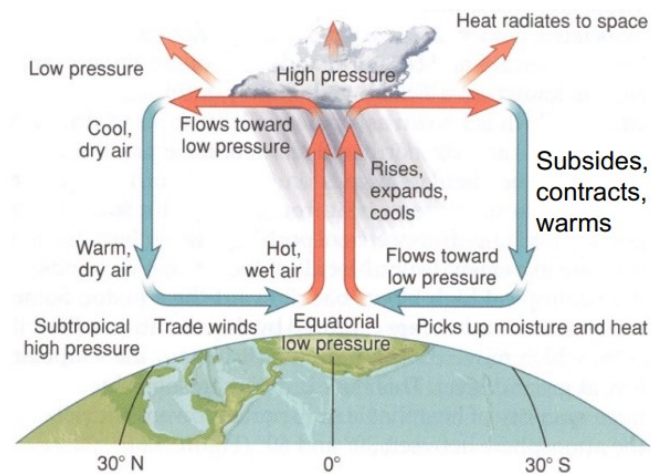


Average troposphere ($\Gamma_{env} = 6.5\text{Ckm}^{-1}$) is nearly neutral to moist convection.

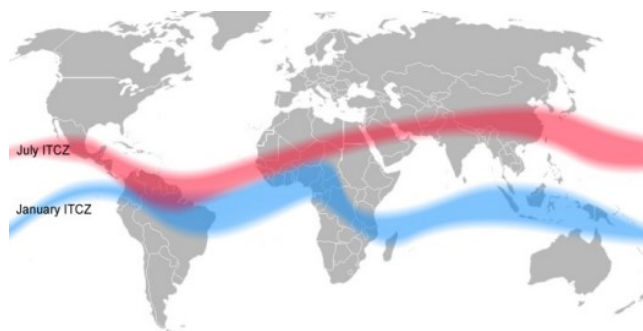
- Equivalent Potential Temperature

$$\theta_e = \theta \exp\left(\frac{Lq}{c_p T}\right)$$

Hadley Circulation in Tropics

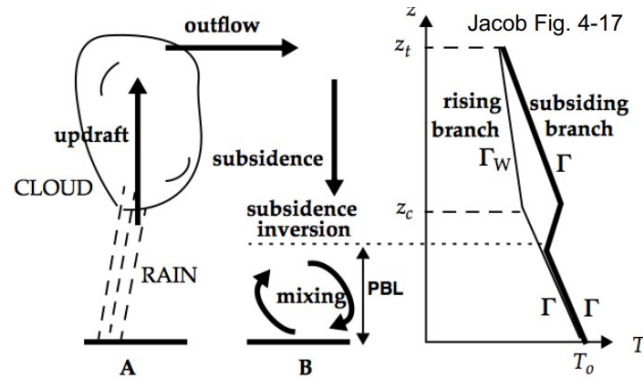


- Deep convection up to the tropopause near the equator, where solar heating is the strongest.
- ITCZ (Intertropical Convergence Zone) varies seasonally
- Subsidence branch of the Hadley cell is a major reason why the world's deserts are mostly located around 30N or 30S.



Moist Convection and Subsidence Inversion

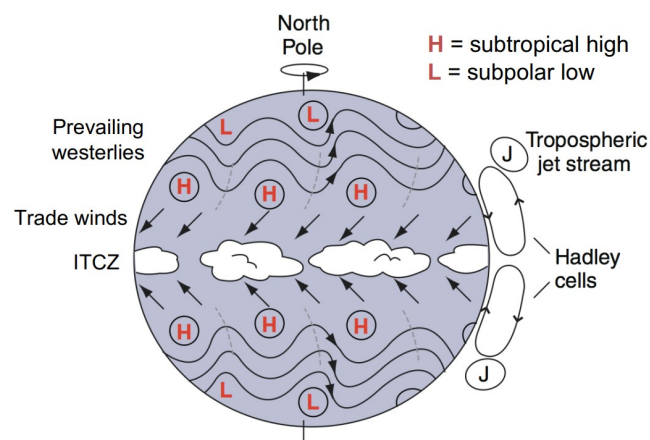
- As the dry, warm sinking air meets colder transient convective air from the surface, an inversion layer known as subsidence inversion is formed and caps the mixed layer.



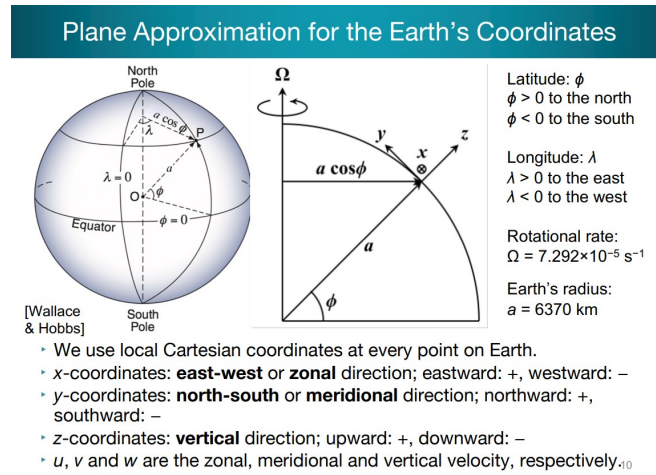
Types of Precipitation

- Frontal: Two air masses of different temperatures come into contact
- Convective: Adjacent layers of moist air are uplifted due to excessive heating
- Organographic Mountain force air masses to be uplifted until saturation is reached on the windward side

General Circulation of Atmosphere



Lecture 9 - Angular Momentum



$$L = m\Omega(a \cos \phi)^2 + mua \cos \phi$$

Lecture 10 - Geostrophic Balance

Composition of Forces on Earth

- Real forces
 - Pressure gradient force
 - Gravitational force
 - Frictional force
- Apparent forces
 - Centrifugal force
 - Coriolis force

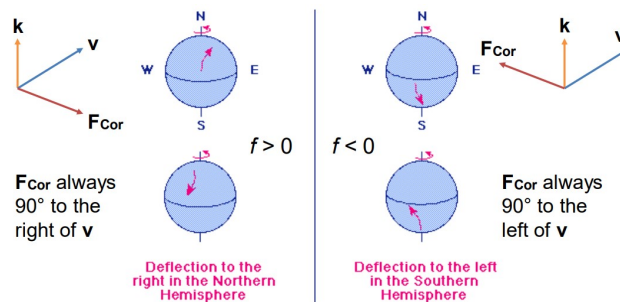
Coriolis Force

$$F_{cor,x} = m \frac{du}{dt} = f v = 2v\Omega \sin \phi$$

$$F_{cor,y} = m \frac{dv}{dt} = -f u = -2u\Omega \sin \phi$$

$$\mathbf{F}_{Cor} = -f \mathbf{k} \times \mathbf{v}$$

- In the Northern Hemisphere ($f > 0$), as an air parcel moves northward ($v > 0$), its u increases (or becomes more positive). The air parcel is deflected to the right! If it moves southward ($v < 0$), u decreases (or becomes more negative); it is still deflected to the right!
- In the Southern Hemisphere ($f < 0$), a meridionally moving air parcel is deflected to the left.



Pressure Gradient Force

Consider an air parcel with density ρ and volume $A\Delta x$, net force per unit mass in x-direction is $F_{net,x} = \frac{p_1 A - p_2 A}{\rho A \Delta x}$.

Pressure gradient force: $\mathbf{F}_{PG} = -\frac{1}{\rho}\nabla p$, air flows from **high to low** pressure.

In z-direction, $F_{net,z} = 0$ is the hydrostatic balance, $F_{net,z} = -\frac{1}{\rho}\frac{\partial p}{\partial z} - g$.

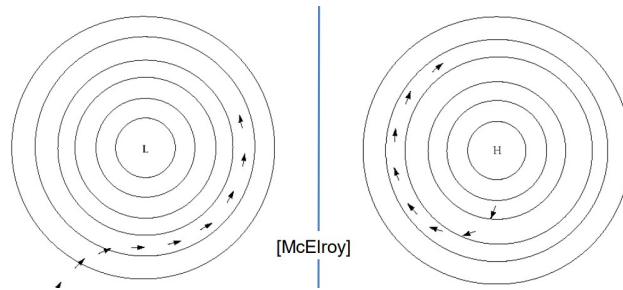
Horizontal Momentum Equation & Geostrophic Balance

$$\frac{dv}{dt} = \mathbf{F}_{PG} + \mathbf{F}_{Cor} + \mathbf{F}_{fric} = -\frac{1}{\rho}\nabla p - f\mathbf{k} \times \mathbf{v} + \mathbf{F}_{fric}.$$

For synoptic large-scale (≥ 1000 km) atmospheric circulation patterns from middle to high latitudes in the free troposphere, $\mathbf{F}_{PG}$ and $\mathbf{F}_{Cor}$ become the dominant terms in the momentum equation.

$$\mathbf{v}_g = \frac{1}{\rho f}\mathbf{k} \times \nabla p.$$

In the Northern Hemisphere, wind flows counterclockwise around a low pressure, but clockwise around a high. Opposite in the Southern Hemisphere.



Lecture 11 - Thermal Wind

Cyclonic and Anticyclonic

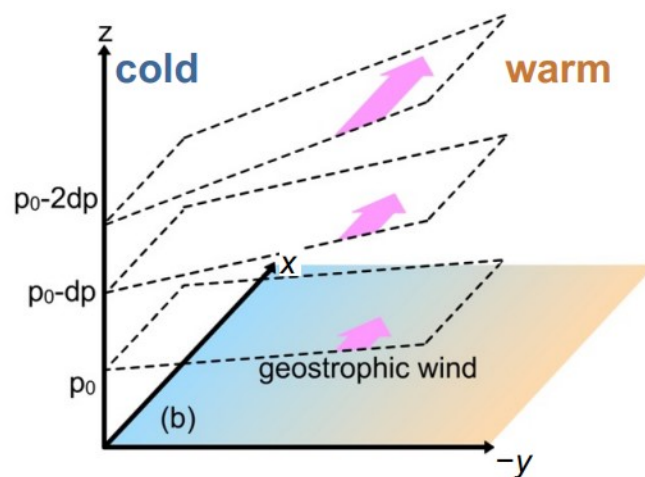
	<i>Northern Hemisphere</i>	<i>Southern Hemisphere</i>
Cyclonic =	Counterclockwise 	Clockwise 
Anticyclonic =	Clockwise 	Counterclockwise 

- Flows around a low pressure are always cyclonic
- Flows around a high pressure are always anticyclonic

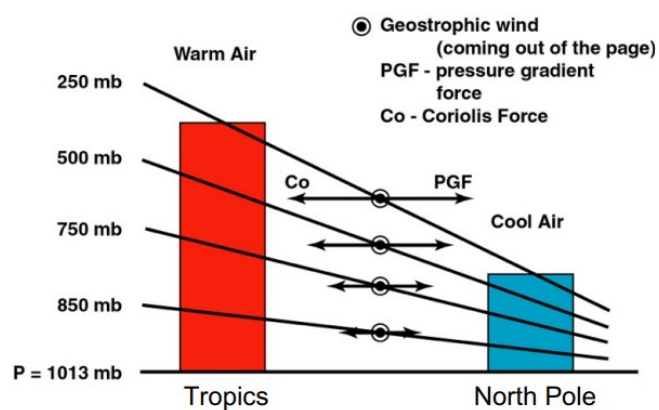
Geostrophic Winds in Pressure Coordinates

$$\mathbf{v}_g = \frac{g}{f} \mathbf{k} \times \nabla Z$$

Thermal Wind Balance



Temperature slopes should become steeper with height, meaning that the associated geostrophic wind should also increase with height.



Strong meridional (equator-to-pole) temperature gradient can lead to strong vertical gradient of zonal wind.

$$\frac{\Delta u_g}{\Delta Z} = -\frac{g}{f\bar{T}} \frac{\partial \bar{T}}{\partial y}.$$

Strong meridional (equator-to-pole) temperature gradient leads to strong vertical wind shear and thus strong upperlevel zonal winds → jet stream



Frictional Force

$$F_{fric,x} = \nu \frac{\partial^2 u}{\partial z^2}, F_{fric,y} = \nu \frac{\partial^2 v}{\partial z^2}$$

In the free troposphere above the boundary layer (typically 1–2 km thick), F_{fric} is usually negligible. Within the boundary layer, F_{fric} must be taken into account. The layer in which F_{fric} becomes important is called the Ekman layer, characterized by a thickness δ .

$$\mathbf{F}_{fric} = -\frac{k}{\delta} \mathbf{v}, k = \text{drag coefficient due to surface roughness.}$$

- $\mathbf{F}_{PG}$ same as before
- $\mathbf{F}_{Cor}$ and $\mathbf{F}_{fric}$ together balance $\mathbf{F}_{PG}$
- $|\mathbf{F}_{Cor}| < |\mathbf{F}_{PG}|$
- $\mathbf{F}_{fric}$ opposite to flow ($\mathbf{v}$) and $\mathbf{F}_{Cor}$ still perpendicular to flow

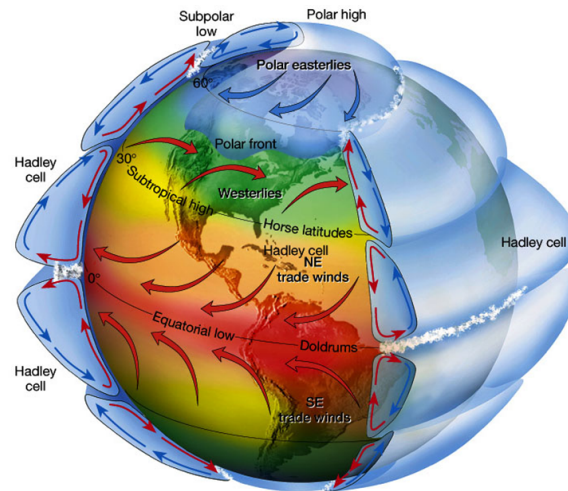
Effect of frictional drag is that:

- It slows down the flow (making $|\mathbf{v}| < |\mathbf{v}_g|$ → **subgeostrophic flow**);
- It creates a cross-isobar flow ($\mathbf{v}_{ag} = \mathbf{v} - \mathbf{v}_g$) toward lower pressure → **ageostrophic flow**

[Wallace & Hobbs Ch.7]

actual flow ($\mathbf{v}$) =
 geostrophic flow ($\mathbf{v}_g$) +
 ageostrophic flow ($\mathbf{v}_{ag}$)

Lecture 12 - General Circulation

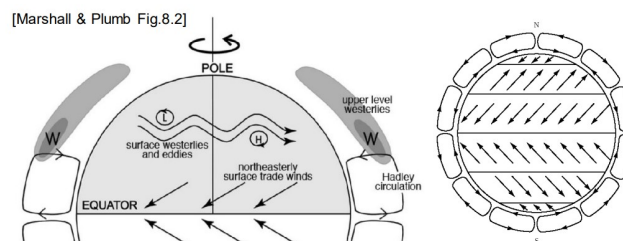


A hemisphere-wide overturning circulation is observed on Venus, but not on Earth due to strong rotational constraint.

Hadley Circulation in the Tropics

As air in the rising branch of the Hadley cell reaches the tropopause, it cannot penetrate and thus is forced to move poleward.

By conservation of angular momentum, the poleward-moving upper-level air gains an increasing eastward velocity, creating strong westerly winds aloft. This is consistent with the observation of the subtropical jet, which is, of course, also consistent with thermal wind balance.



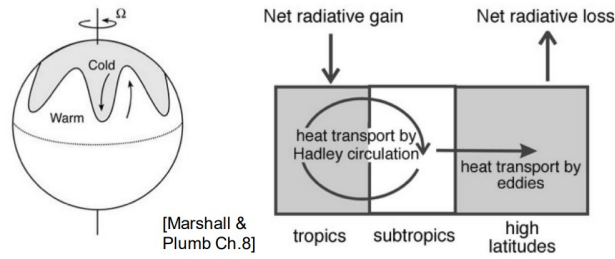
Equatorward branch at the surface is deflected by Coriolis force to the west, forming easterly winds known as trade winds.

Subsiding air at 30° latitude at the surface flows poleward, and is deflected by Coriolis force to the east, forming the prevailing westerlies.

Earth's strong rotational constraint that leads to high instability of the atmosphere, leading to the formation of eddies (wave-like, longitudinally varying flow).

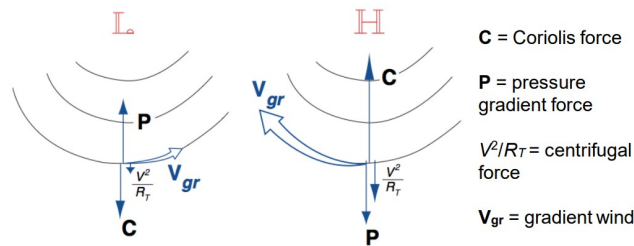
Conservation of angular momentum in the upper-level poleward branch of the Hadley cell would lead to very strong zonal flow aloft with large vertical wind shear → hydrodynamically unstable

Poleward Transport of Energy



- Cold (usually continental) and warm (usually maritime) air masses are mixed by baroclinic eddies.
- Baroclinic eddies are largely responsible for poleward heat transport from middle to high latitudes.

Gradient Wind Balance



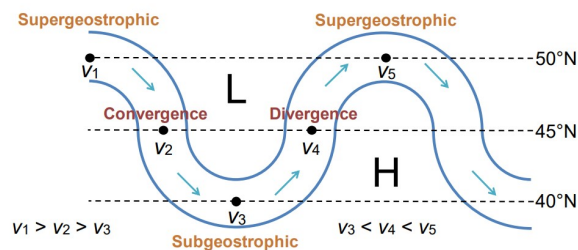
- Even at upper levels, sharp curvature of flow would introduce centrifugal force that causes wind to depart from geostrophy. The resulting actual wind is called gradient wind.
- Around a low, $|\mathbf{v}_{gr}| < |\mathbf{v}_g| \rightarrow$ **subgeostrophic**,

$$|f|v + \frac{v^2}{R} = g \left| \frac{\Delta Z}{\Delta r} \right|$$

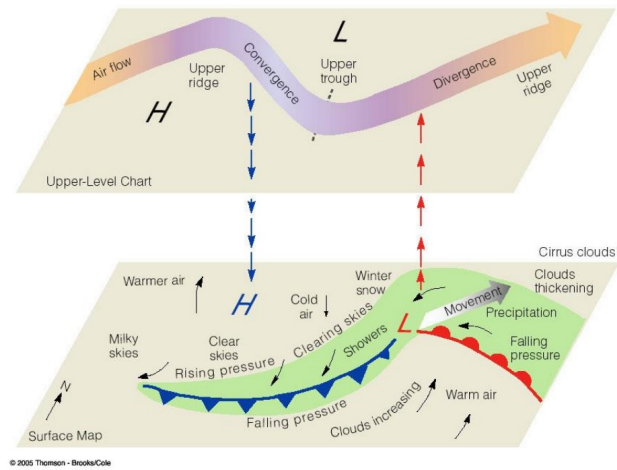
- Around a high, $|\mathbf{v}_{gr}| > |\mathbf{v}_g| \rightarrow$ **supergeostrophic**,

$$|f|v = \frac{v^2}{R} + g \left| \frac{\Delta Z}{\Delta r} \right|$$

For the upper-level winds flowing wave-like around low-pressure troughs and high-pressure ridges, gradient winds result in divergence (convergence) of air east (west) of the low, inducing vertical motion of air from (to) the surface.



Formation of Midlatitude Cyclones and Fronts



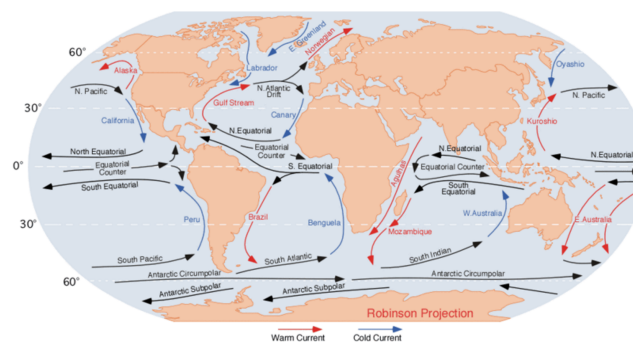
Lecture 15 - Ekman Transport

Latitudinal Imbalance of Radiant Energy

- The atmosphere is responsible for 2/3 of the poleward heat transport, mainly by the Hadley circulation in the tropics and baroclinic eddies at higher latitudes.
- The ocean is responsible for the other 1/3, mainly by a combination of wind-driven circulation and thermohaline (buoyancy-driven).

Vertical Structure of the World's Oceans

- Density gradient ((**thermocline**)) divides oceans into upper surface or mixed layer and the lower deep or abyssal layer.
- Deep convection makes the polar oceans relatively homogeneous throughout their depths.



Characterized by major ocean gyres centered around the subtropics (clockwise in N. Hem. and counterclockwise in S. Hem.); smaller gyres in the subpolar and tropical regions.

Geostrophic Balance of Ocean Circulation

- Water is almost incompressible
- Oceans are mostly bounded by the continents
- Convection is forced by cooling at the sea surface (“cooling from above”)
- Wind may drive the ocean circulation while the ocean seldom drives the wind

The convection in the atmosphere is driven by differential heating, while the deep water formation in the ocean is mainly caused by cooling.

Geostrophic Currents

-

$$\mathbf{v}_g = \frac{g}{f} \mathbf{k} \times \nabla h$$

- **Thermal** sea surface height expansion: subtropical high pressure systems associated with clear skies $\rightarrow$ solar heating $\uparrow \rightarrow$ thermal expansion of surface ocean layer $\rightarrow$ ocean surface height $\uparrow$
- **Dynamical** sea surface height difference: Ekman transport related convergence at the center of the subtropical ocean gyre, which increases the sea surface height at the center of a subtropical gyre

$$|\mathbf{M}_{Ek}| = \left| \frac{\tau_{wind}}{f} \right|,$$

mass transport to the right (left) of the surface wind in the Northern (Southern) Hemisphere.

Effect of Ekman Transport

- In the Northern Hemisphere,
 - anticyclonic wind applied stress will cause convergence and downwelling (subtropics)
 - cyclonic wind applied stress will cause divergence and upwelling (tropics and subpolar regions)
- **Subpolar upwelling** brings nutrients from the deep water $\rightarrow$ supports biological activities $\rightarrow$ high chlorophyll concentration
- **Subtropical downwelling** removes nutrients from the surface layer $\rightarrow$ suppresses biological activities $\rightarrow$ "biological deserts"

Lecture 16 - Thermohaline Circulation

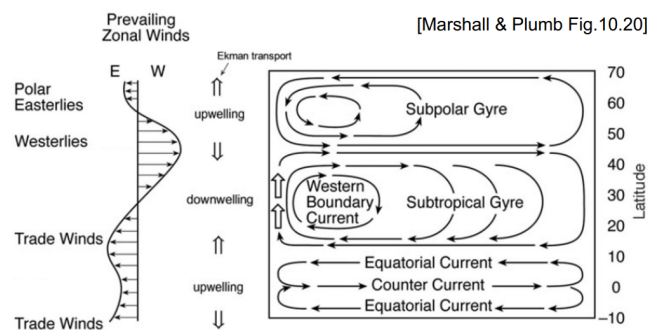
Ekman Pumping and Response of Interior Ocean

Average meridional current in the ocean basin:

$$V \approx \frac{f w_{Ek}}{\beta D}.$$

Therefore, the **downwelling** of the **subtropical gyre** induces an overall **equatorward flow** in the ocean basin. The subtropical atmosphere imparts anticyclonic rotation to the ocean. The coastal friction must impart cyclonic rotation on the poleward return flow, and is only achievable on the **western boundary**.

- Gulf stream
- Kuroshio current (Poleward)
- Oyashio current (Equatorward)



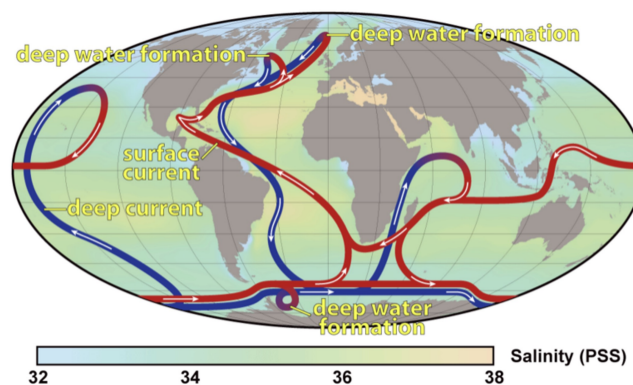
Physical Mechanism of Deep Water Formation

Density increase of sea water due to:

- Intense surface cooling
- Formation of the sea ice which excludes salt → increases the salinity of the surrounding water → denser

It occurs in the Northern Atlantic and in the Southern Ocean around Antarctica.

Thermohaline Circulation



Deep water originated in the North Atlantic flows south, circling Antarctica picking up additional contribution from the Southern Ocean, and enters the Indian and Pacific Oceans where it slowly and diffusely returns to the surface and finally returns to the Atlantic.

In each ocean, the subtropic is more saline than higher latitudes since evaporation exceeds precipitation and runoff from rivers in the subtropics. Excess water vapor is partly transported poleward contributing to an excess of precipitation over evaporation at higher latitudes.

- Why is deep water not formed in the North Pacific?

This is because Atlantic Ocean is generally more saline than the Pacific Ocean. The water vapor accompanied by prevailing westerlies in the midlatitude Pacific is blocked by the Rocky Mountains in North America and the Andes in the South America, whereas easterly trade winds in the tropics across Central America encounters fewer geographical barriers and there is a net transport of water vapor from the Atlantic to the Pacific.

Meridional Overturning Circulation (MOC)

- There is a global-scale thermohaline circulation due to the cold, saline seawater in the North Atlantic and Southern Ocean in the Antarctic region causing deep water convection and sink, there must be a large-scale upwelling of deep water back to the surface oceans.
- There is a net poleward heat transport by MOC. Without the ocean heat transport, the higher latitudes will be colder, and also intensified by the ice-albedo feedback.
- Global warming might freshen seawater and weaken or even shutdown the MOC, it might partly offset the warming in the North Atlantic. However, it is likely that the North Atlantic will still get warmer, but with a slower rate comparing to the rest of the Northern Hemisphere. (Less dense and less likely to sink)
 - Higher precipitation at higher latitudes
 - Melting of sea ice and land ice sheets
 - Warming of seawater

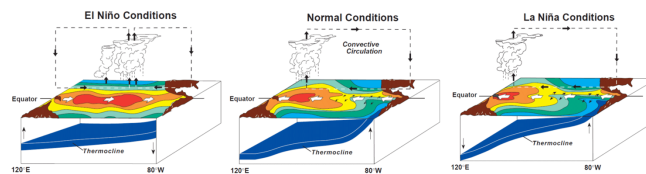
Lecture 17 - Ocean and Climate

The Ability of Ocean to Moderate Energy Balance

- **Specific heat capacity:** Water has a specific heat about four times higher than that of dry land.
- **Effective depth:** Turbulent mixing of the surface ocean allows heat to penetrate much deeper in the ocean than on land, where heat can only penetrate by conduction.
- Ocean can reduce the **diurnal and seasonal fluctuations** in temperature

El Niño-Southern Oscillation (ENSO)

- A climate phenomenon involving the atmosphere-ocean system over **equatorial Pacific**.
 - El Niño (warm phase): anomalous warming of the sea off Peruvian coast
 - La Niña (cold phase): anomalous cooling of the sea off Peruvian coast



Normal Condition

Easterly trade winds pile up warm water in the west, and cause upwelling in the east that brings cold water from the deep → sea surface is colder and thermocline is shallower in the eastern equatorial Pacific

- Convection in the west and sinking in the east (Walker circulation)
- Strengthens the trade winds

El Niño

Trade winds weaken and cause anomalous warming in the east → the weakening also suppresses upwelling → further warming in the east and flattening of thermocline across equatorial Pacific → convective region moves eastward

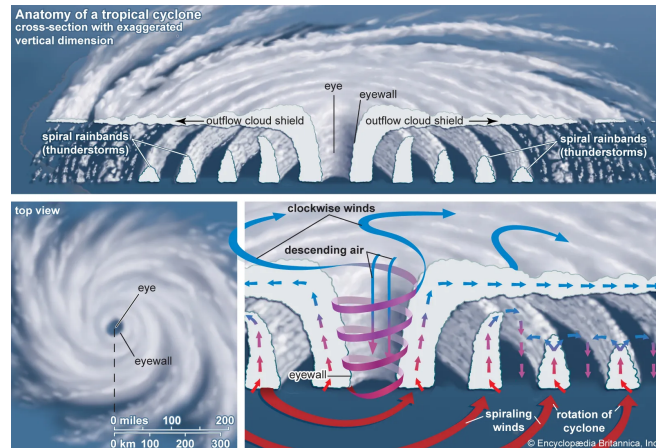
- Warmer weather in the entire eastern Pacific
- Decline in fisheries off the west coast of South America
- Rainfall is reduced in the western tropical Pacific but enhanced in the eastern tropical Pacific e.g., drought in Indonesia but flooding near Peru

La Niña

Trade winds strengthen and cause anomalous cooling in the east → the strengthening also reinforces upwelling → further cooling in the east and a more tilted thermocline across equatorial Pacific → convective region moves westward

Tropical Cyclones

- Formed **bottom-up**
 - Water temperature $\geq 26.5^{\circ}\text{C}$
 - Rapid cooling with height in the atmosphere (stable)
 - High humidity
 - Small vertical wind shear
 - More than 5 latitude away from the equator
 - Pre-existing disturbance



Cyclostrophic Balance

$$\frac{v^2}{r} = \frac{1}{\rho} \frac{\partial p}{\partial r} \Rightarrow v = \sqrt{\frac{r}{\rho} \frac{\partial p}{\partial r}}$$

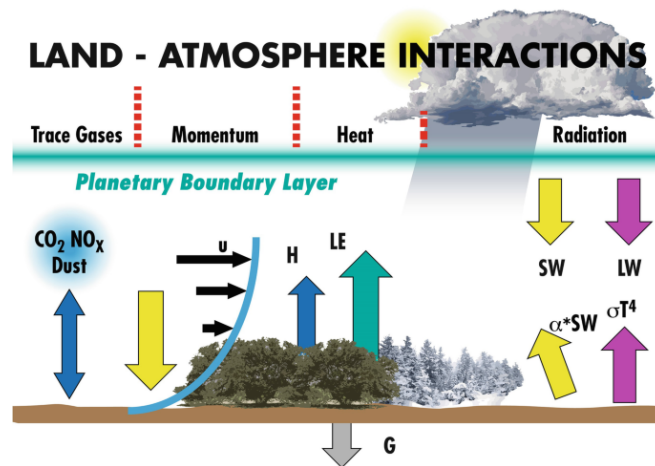
At the surface, friction induces a radially inward wind component leading to a convergence of air toward the cyclone eye.

- Storm surge
 - **Wind blowing** across the surface pushes water onshore by Ekman Transport
 - Pressure-induced water suction

Dissipation of Tropical Cyclones

1. Drifting to higher latitudes: Sea surface temperature is no longer be warm enough to sustain the cyclone, and stronger vertical wind shear could tear it apart.
2. Landfall: Absense of latent heat flux and friction
3. Drifting to midlatitudes: Encountering an extratropical cyclone along the storm track and intensify, forming a superstorm

Lecture 18 - Biogeophysical Fluxes



Biogeophysical processes include the exchange of momentum, water and energy (heat + radiation) between the surface (soil, vegetation, water body) and atmosphere.

Momentum

Roughness of the surface imposes a drag or friction on the atmosphere, which in turn imposes a stress (τ) on the surface, shaping wind velocities (u) in the boundary layer and creating vertical wind shear (gradient).

- Wind shear and surface heating lead to dynamic and static instability, generating **mechanical** and **thermal** turbulence, respectively.
- Turbulent mixing is an important driver of land-atmosphere exchange of heat, water and chemical substances, shaping climate and atmospheric composition at large.

Heat and Radiation

- H = sensible heat flux (conduction and convection to air)
- λE = latent heat flux (λ : specific heat of vaporization of water; E : evapotranspiration)
- G = ground heat flux (conduction to ground)
- S = shortwave radiation
- L = longwave radiation
- α = surface albedo

Surface Energy Balance

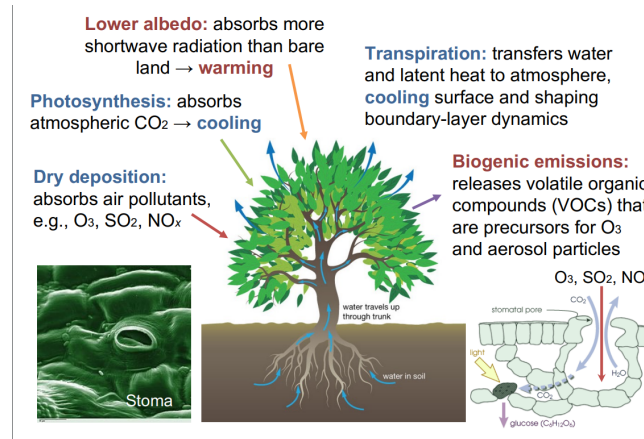
$$R_n = (1 - \alpha)S + (L - \sigma T_s^4) = H + \lambda E + G$$

Evapotranspiration

- Sum of surface **evaporation** (from soils or water bodies) and plant **transpiration** (transport of water from plant roots to leaves and release from stomata).
- Releases latent heat and warms upper air → **cooling effect**
- Transpiration transfers water and heat to atmosphere, and accounts for 80–90

Albedo of Land Plants

- Brighter objects have higher albedo (reflect more visible light)
- Vegetation is generally less reflective of solar shortwave radiation than bare lands, thus when only radiative balance is concerned it has a warming effect.
- Albedo: boreal needleleaf ; temperate/tropical broadleaf ; grass



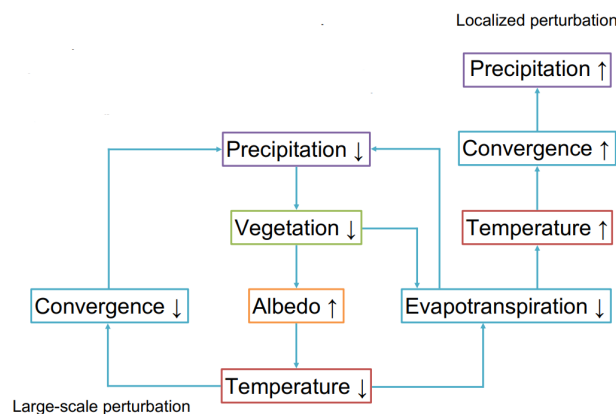
- Vegetation generally increases evapotranspiration and latent heat flux.
- **Boreal forests** (conifers) mainly have a **warming effect**, due to **low albedo** and **limited transpiration of their leaf properties**. Its expansion with ice-albedo feedback further amplifies warming in the Arctic
- **Temperate forests** has **low albedo** and its **transpiration is limited by soil-water availability** and partly offset the CO₂ uptake
- **Tropical broadleaf trees** has **higher albedo**, **transpiration is not limited by water availability** and has **strong carbon uptake**. **Reforestation of broadleaf trees in the tropics** can cause a **strong cooling effect**.

$T \uparrow \rightarrow \text{photosynthesis} \uparrow \rightarrow \text{albedo} \downarrow \rightarrow T \uparrow$

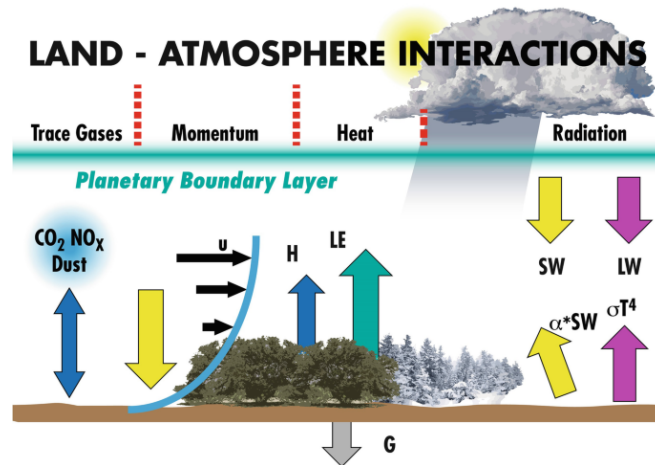
$T \uparrow \rightarrow \text{photosynthesis} \downarrow \rightarrow \text{albedo} \uparrow \rightarrow T \downarrow$

$T \uparrow \rightarrow \text{photosynthesis} \uparrow \rightarrow \text{evapotranspiration} \uparrow \rightarrow T \downarrow$

$T \uparrow \rightarrow \text{photosynthesis} \downarrow \rightarrow \text{evapotranspiration} \downarrow \rightarrow T \uparrow$



Lecture 19 - Biogeochemical Fluxes



Biogeochemical processes include the exchange of chemical substances (esp. of C, N, S and P) between the surface and atmosphere

Rock Cycle

- **Weathering:** physical and chemical breakdown of rocks and minerals by heat, water, ice, pressure, air, wind, living organisms, etc.
- **Erosion:** removal and transport of weathered materials to another location mainly by water flow and wind.
- **Sedimentation:** deposition of weathered materials onto the ocean bottom or land surface mainly by gravity.
- **Subduction:** one tectonic plate moving underneath another plate at a convergent boundary, sinking into the mantle.
- **Metamorphism:** physical and chemical change of rocks due to exposure to heat and pressure beneath the surface.
- **Volcanism:** eruption of molten rock (magma) onto the surface.

Geological "Slow" Carbon Cycle

Could take $10^6 - 10^9$ years

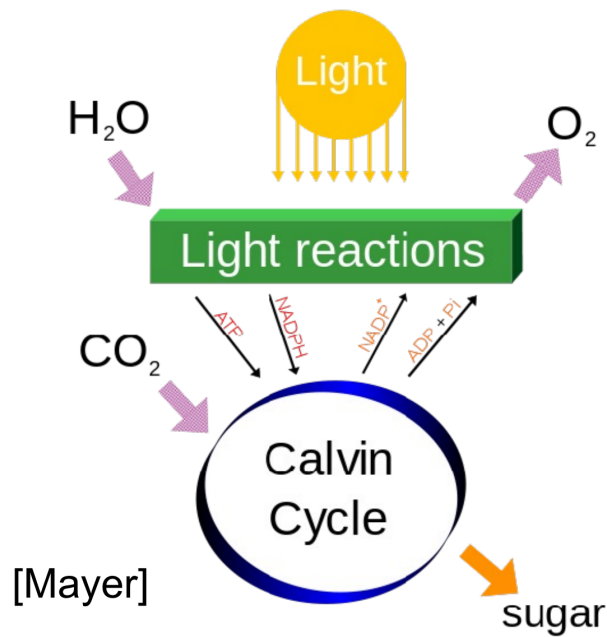
Weathering, erosion and sedimentation: $\text{CaSiO}_3 + \text{CO}_2 \rightarrow \text{CaCO}_3 + \text{SiO}_2$

Subduction, metamorphism and volcanism: $\text{CaCO}_3 + \text{SiO}_2 \rightarrow \text{CaSiO}_3 + \text{CO}_2$

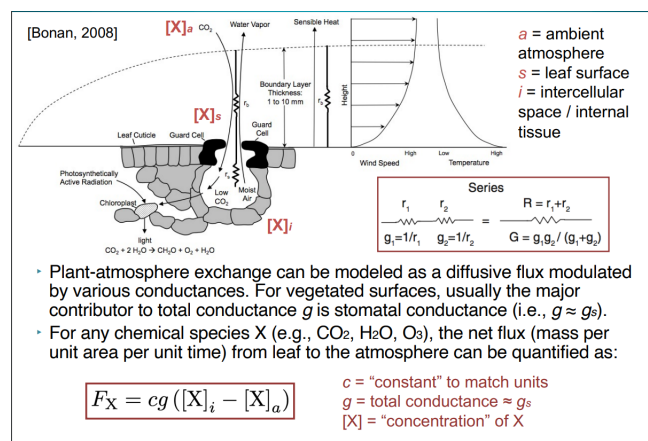
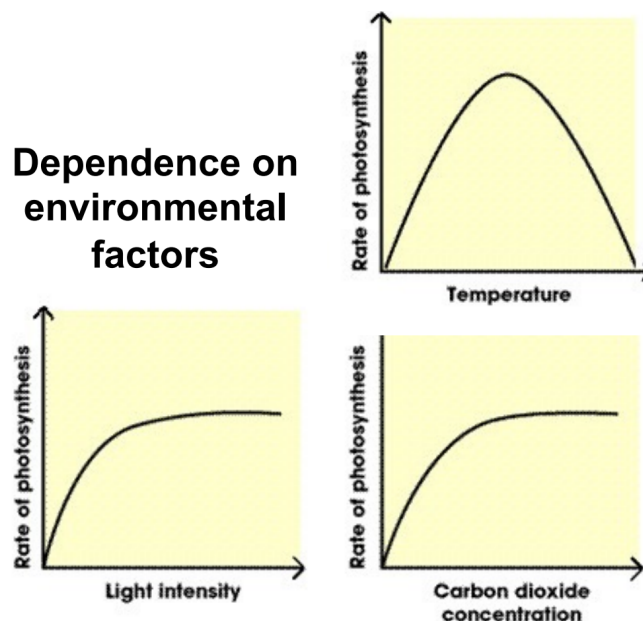
Self-regulating feedback: higher $\text{CO}_2 \rightarrow$ warmer surface $\rightarrow$ faster rate of weathering $\rightarrow$ more CO_2 uptake $\rightarrow$ lower CO_2

Photosynthesis

- Absorbs atmospheric CO_2 and combines with H_2O to make food $\rightarrow$ **cooling effect**
- Energy from sunlight is converted and stored in reduced carbon compounds (mostly as carbohydrates and lipids).
- $\text{CO}_2 + \text{H}_2\text{O} + h\nu \rightarrow \text{CH}_2\text{O} + \text{O}_2$



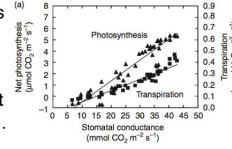
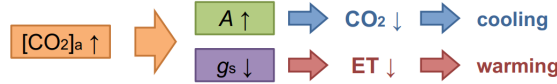
Stomata open to allow CO₂ uptake for photosynthesis, but close to prevent excess water loss from transpiration. Stomata can maximize photosynthetic CO₂ gain and minimize transpirational water loss.



Total photosynthetic CO₂ flux: $A = -F_{\text{CO}_2} \approx -c g_s ([\text{CO}_2]_i - [\text{CO}_2]_a)$

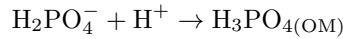
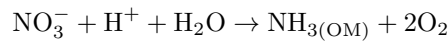
Total transpirational water flux: $E = F_{\text{H}_2\text{O}} \approx c g_s [e_s(T_i) - e_a]$

- Stomatal conductance (g_s) is regulated so as to maximum photosynthetic CO₂ gain (A) and to minimize transpirational water loss (E), resulting in an almost linear positive relationship between g_s and A for a given set of conditions (e.g., for a given e_a and $[\text{CO}_2]_a$).
- Dependence of g_s on environmental conditions is complex. Generally it increases with $A(h\nu, T, [\text{CO}_2]_a, \dots)$ and e_a , but decreases with increasing $[\text{CO}_2]_a$, reflecting the optimization between A and E .
- Overall effect of atmospheric CO₂ increase:



Nutrient Uptake and Assimilation

- Nitrate and phosphate assimilation (NO_3^- , PO_4^{3-})

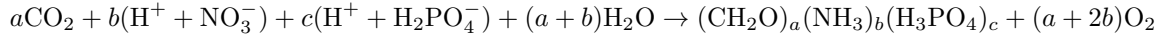


- Ions from soil solution

By mineral rocks weathering

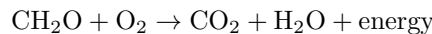
Ultimate Source of N is the atmosphere through **nitrogen fixation**

- Terrestrial **primary production**

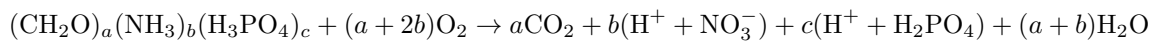


C:N:P ratio for land plants on average is 882 : 9 : 1.

Respiration



- Reduced carbon is oxidized in cellular respiration to release energy (in form of ATPs) for a variety of uses.
- Some microbes can perform anaerobic respiration in anoxic conditions
- During decomposition of dead organic matter by microbes, organic (reduced) N and P are released back to environment via mineralization:



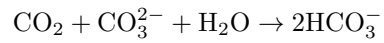
- From organic N to NH_4^+ to NO_3^- : **ammonification and nitrification**

Litter and Humus

- Litter: undecomposed materials (e.g. leaves, twigs, branches, carcasses)
- Humus: layer underneath formed by highly decomposed organic materials
- Organic matter remaining can be expressed as $x = x_0 e^{-kt}$, where k is the decay rate

Ocean Biogeochemical Processes

Dissolution of Atmospheric CO₂



Marine phytoplankton:

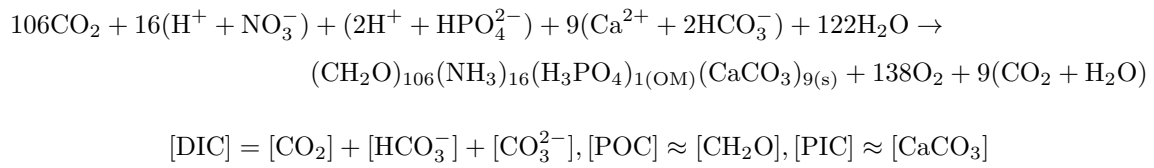
Calcite formation:



Redfield ratio: C:N:P $\approx$ 106 : 16 : 1 is relative constant.

Phytoplankton as a whole forms particulate **organic carbon** (POC) and particulate **inorganic carbon** (PIC) from dissolved inorganic carbon (DIC) in a ratio of roughly **12:1**.

Ocean **primary production:**



Lecture 20 - Carbon Cycle I

Reservoir and Residence Time

$$\tau = \frac{M}{F},$$

where M is the total stock (mass or volume) in a reservoir, F is the flux into or out of the reservoir
In steady state,

$$\sum F_{in} = \sum F_{out}$$

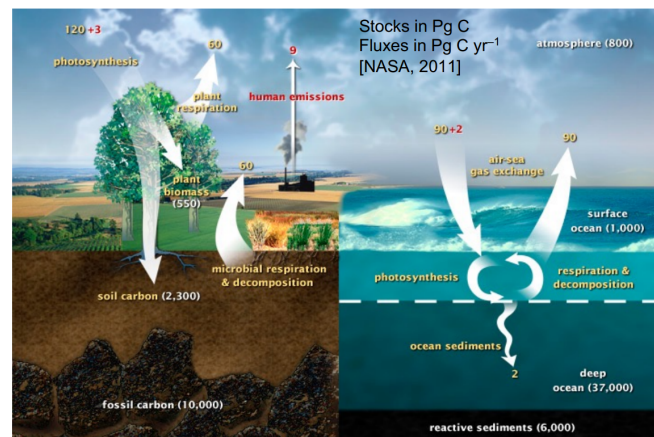
Mass unit conversion:

$$1 \text{ Gt} = 10^9 \text{ ton} = 10^{12} \text{ kg} = 10^3 \text{ Tg} = 1 \text{ Pg}$$

Observed Atmospheric CO₂ Concentration

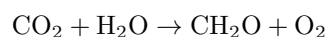
- Seasonal cycle: photosynthesis as sink from spring to summer, and net respiration from fall to winter

Contemporary "Fast" Carbon Cycle



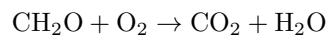
- Photosynthesis

—



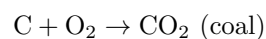
- Respiration and Decomposition (and Deforestation)

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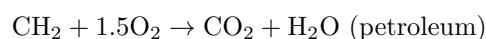


- Fossil fuel combustion

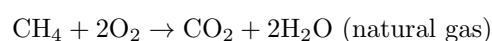
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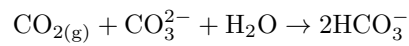


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- Ocean uptake:

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- Calcite formation:

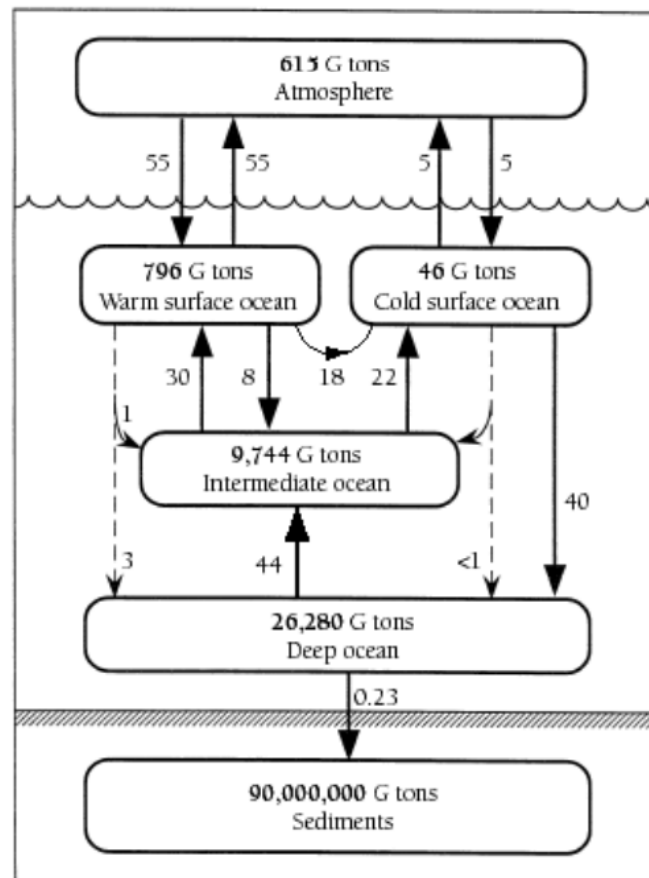
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Global Anthropogenic CO₂ Emissions

- 2.13 Pg of carbon emission into the atmosphere is equivalent to increasing CO₂ level by 1 ppm.
- Only roughly half of CO₂ emitted remained in the atmosphere. The other half was taken up together by the oceans and the terrestrial biosphere-soil system.

Pre-industrial Terrestrial Carbon Cycle



- Most of the carbon stock is in the form of inorganic bicarbonate ions.
- Marine phytoplankton build calcium carbonate shells (CaCO₃) while photosynthesizing.
- Carbon lifetime in atmosphere-biosphere-soil-ocean system is 170,000 yrs, the carbon lifetime in sedimentary reservoir is 390 million years

Perturbation of Global Carbon Cycle

- Any perturbation can be partly buffered by terrestrial-ocean system by reforestation and biological pump, but only for tens to hundreds of years.
- Fossil fuel combustion greatly accelerates the carbon return rate from geological reservoir to atmosphere. However, it takes > 100000 years to get out of the atmosphere-terrestrial-ocean system.

Marine Carbon Cycle: Geochemical Aspects

•

$$\text{CO}_{2(\text{g})} \rightleftharpoons \text{CO}_{2(\text{aq})}, K_H = \frac{[\text{CO}_{2(\text{aq})}]}{p_{\text{CO}_2}}$$

•

$$\text{CO}_{2(\text{aq})} + \text{H}_2\text{O} \rightleftharpoons \text{H}^+ + \text{HCO}_3^-, K_{a1} = \frac{[\text{H}^+][\text{HCO}_3^-]}{[\text{CO}_{2(\text{aq})}]}$$

•

$$\text{HCO}_3^- \rightleftharpoons \text{H}^+ + \text{CO}_3^{2-}, K_{a2} = \frac{[\text{H}^+][\text{CO}_3^{2-}]}{[\text{HCO}_3^-]}$$

- Net overall reaction:

$$\text{CO}_{2(\text{g})} + \text{CO}_3^{2-} + \text{H}_2\text{O} \rightleftharpoons 2\text{HCO}_3^-, K' = \frac{K_H K_{a1}}{K_{a2}}$$

•

$$\text{Alk} = [\text{HCO}_3^-] + 2[\text{CO}_3^{2-}] + [\text{B}(\text{OH})^4] + [\text{OH}^-] - [\text{H}^+],$$

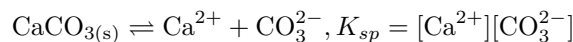
$$\text{Alk} \approx [\text{HCO}_3^-] + 2[\text{CO}_3^{2-}] \text{ (in the ocean)}$$

. Addition or removal of weak acids does not change the alkalinity of a solution. Addition (removal) of strong acids decreases (increases) alkalinity.

- Total **dissolved inorganic carbon** (DIC or $\sum \text{CO}_2$):

$$[\text{DIC}] = [\text{CO}_{2(\text{aq})}] + [\text{HCO}_3^-] + [\text{CO}_3^{2-}].$$

- Typical ocean pH: 8.2, most of DIC is in bicarbonate ion forms.
- Warming limits ocean's ability to dissolve atmospheric $\text{CO}_2 \rightarrow$ positive feedback
- Calcium carbonate mineral in suspended particles, sediments and shells:



- When $[\text{CO}_3^{2-}]$ increases, p_{CO_2} at the sea surface would decrease, there is a net CO_2 flux into the ocean.

Lecture 22 - Carbon Cycle II

Marine Carbon Cycle - Biochemical Aspects

- Solubility pump: Dissolved inorganic carbon (DOC) and inorganic carbonate species are "pumped" down the deep ocean by downwelling of cold, dense water in polar regions.
- Biological pump: Dead bodies and fecal pellets sink as particulate organic carbon (POC) together with CaCO_3 shells to the deep ocean, effectively transfers carbon from the atmosphere to deep ocean and lithosphere.

O₂ and CO₂ Ratio

- 2.13 PgC of carbon emissions into the atmosphere is equivalent to increasing CO₂ level by 1 ppm.
- Fossil fuel combustion on average consumes 1.4 mol of O₂ per mol of CO₂ released.
- Photosynthesis on average releases 1.1 mol of O₂ per mol of CO₂ absorbed.
- Respiration and decomposition on average consumes 1.1 mol of O₂ per mol of CO₂ released.
- A change in O₂ mixing ratio of 1 ppm is equivalent to a change in $\delta(\text{O}_2/\text{N}_2)$ of 4.77 per meg.
- Ocean uptake or release of CO₂ does not consume or release O₂.

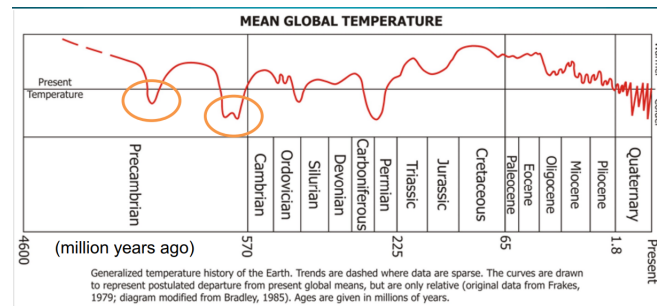
Effects of Warming on Soil and Vegetation

- Soil
 - Decomposition $\uparrow \rightarrow$ soil C loss
 - N mineralization $\uparrow \rightarrow$ vegetation C loss
- Vegetation
 - $T \uparrow +$ longer growing season $\rightarrow$ NPP $\uparrow$ (given enough water)
 - $T \uparrow +$ water stress $\uparrow \rightarrow$ NPP $\downarrow$ (given insufficient water)
- Indirect effects (leading to C loss)
 - Extreme events (drought, heat wave, hurricane) $\uparrow$
 - Thawing of permafrost
 - Wildfires and insect outbreaks $\uparrow$

Biogeochemical Feedback Mechanisms

$T \uparrow \rightarrow$ photosynthesis $\uparrow \rightarrow$ CO₂ $\downarrow \rightarrow$ $T \downarrow$
 $T \uparrow \rightarrow$ respiration/decomposition $\uparrow \rightarrow$ CO₂ $\uparrow \rightarrow$ $T \uparrow$
 $T \uparrow \rightarrow$ plant damage $\rightarrow$ photosynthesis $\downarrow \rightarrow$ CO₂ $\uparrow \rightarrow$ $T \uparrow$
 $T \uparrow \rightarrow$ soil mineralization $\uparrow \rightarrow$ plant nutrient $\uparrow \rightarrow$ CO₂ $\downarrow \rightarrow$ $T \downarrow$
 $T \uparrow \rightarrow$ CO₂, CH₄ emissions from wetlands and permafrost $\rightarrow$ $T \uparrow$

Lecture 23 - Climate History



Climate Throughout Earth's History

Earth has hypothetically undergone periods of being completely or nearly completely frozen perhaps three times known as **Snowball Earth**.

Atmosphere of the Early Earth

- Sun was $\sim 30\%$ dimmer than today, however evidence suggests early Earth was warmer than today.
- **Faint Sun Paradox** is resolved mostly by the **high CO_2 concentration** in the early atmosphere.
- Intense volcanism resulted from high energy released by Earth's interior due to **radioactive decay** and **gravitational accretion** of the solar nebula.
- Some H_2O condensed to form **oceans**, which then absorbed CO_2 and sulfur gases from the atmosphere.

Rise of Atmospheric Oxygen

- **Phyotosynthetic bacteria** in the ocean after 3.5 billion years ago led to:
 - Accumulation and rise of oxygen in the **ocean**, and subsequent oxidation of Fe(II) and other reduced mineral ions in the ocean.
 - Accumulation and rise of **atmospheric oxygen**, oxidation of mineral rocks on land and reduced gases in the atmosphere (CH_4).
 - Biological pump and further decline in CO_2
 - Snowball Earth
- **Snowball Earth** by slower plate tectonics and ice-albedo feedback

Getting out of the Snowball Phase

- Oceans frozen $\rightarrow \text{CaCO}_3$ formation and sedimentation stop $\rightarrow$ atmospheric $\text{CO}_2 \uparrow \rightarrow$ surface $T \uparrow \rightarrow$ frozen tropical oceans thaw $\rightarrow$ albedo $\downarrow \rightarrow$ surface $T \uparrow \rightarrow$ ice-albedo feedback $\rightarrow$ "hot house" state
- Rise of Atmospheric Oxygen:
 - Enhanced postglacial flux of phosphate via dust and runoff, accelerated biological pump and caused rise and diversification of animals toward the Cambrian Explosion.
 - Formation of ozone layer shield harmful UV radiation made colonization of land by higher organisms possible

Hot House Phase of Cretaceous

- Ice free, tropical and polar oceans could be as warm as 40°C and 20°C .
- Fossils of tropical plant and animal species can be found at polar latitudes.
- Sea level 100–200 m higher than today!
- Might be attributed to more active volcanism with faster seafloor spreading and plate tectonics.

Long-Term Cooling since Oligocene

- Gradual slowdown of plate tectonics and volcanism lead to decrease in CO_2 level
- Formation of Tibetan plateau and Himalayas after collision of Indian and Eurasian plates $\rightarrow$ land exposed
 $\uparrow \rightarrow$ weathering $\uparrow \rightarrow \text{CO}_2 \downarrow$
- Opening of Drake Passage and closing of Panama Strait $\rightarrow$ formation of Antarctic Circumpolar Current and reorganization of thermohaline circulation $\rightarrow$ reduced poleward heat transport

Ice Ages in Pleistocene

- Ice-albedo feedback
- Milankovitch cycles: **Eccentricity, Precession, Obliquity**
- Lower temperature increases the ocean's ability to dissolve more CO_2 .
- Higher wind speed and higher aridity of the glacial world increase dust influx into the ocean. Additional nutrients accelerate the biological pump

Lecture 24 - Recent Climate

- Temperature Change

Warmer

- Precipitation

Wetter get wetter, drier get drier

- Extreme Events

Increasing warm days and nights

Frequency of heatwaves

Frequency and intensity of heavy precipitation

- Cryosphere

Arctic sea ice extent and Northern Hemisphere spring snow cover is continuing to decrease

Annual mean Antarctic sea ice extent increased

- Ocean

Thermal expansion and level rising

- CO₂ and pH

Ocean acidification

- Calcifying organisms are harmed by lower pH and [CO₃²⁻]

- Biological pump less effective

- Aerosols and clouds

- Cloud albedo effect

- * 1st indirect effect: smaller water droplets (negative forcing)

- * 2nd indirect effect: less precipitation, longer lifetime and water content

- * Semi-direct effect: Black and brown carbon causing warming effect; sulfate, nitrate, organic carbon, dust causing more reflection and cooling effect

High clouds tend to have a warming effect (longwave absorption);

Low clouds tend to have a cooling effect (shortwave reflection)

- Extremely likely that human influence has been the dominant cause of the observed warming since the mid-20th century (fossil fuel combustion)

Lecture 25 - Future Climate

- Representative Concentration Pathway (RCP)
 - scenarios for future emissions or concentrations of anthropogenic greenhouse gases, aerosols and other climate drivers
 - 2.6, 4.5, 6.0, 8.5 identified by their approximate total radiative forcing in year 2100 relative to 1750
 - RCP8.5 no mitigation, business as usual
 - RCP2.6 aggressive mitigation starting right now
- Shared Socioeconomic Pathway (SSP)
 - integrate different sets of population, economic growth, and other socioeconomic assumptions into future emissions scenarios
 - potential no-policy baseline scenarios, as well as how different mitigation scenarios could be attained under different socioeconomic pathways

Projected Physical Changes in the Ocean and Cryosphere

- Global-scale glacier mass loss, permafrost thaw, and decline in snow cover and Arctic sea ice extent

Projected Changes in Temperature, Soil Moisture, and Precipitation

- Temperature increase
- Soil moisture and precipitation: more extreme
- Species loss and human heat shock risks
- Lower food production and fisheries yield

Projections for Tropical Cyclones

- Proportion of intense tropical cyclones and peak wind speeds of most intense tropical cyclones are projected to increase
- Average and maximum rain rates associated will increase
- Sea level rise leads to higher storm surges

Projection for Midlatitude

- Westerly jet stream with uncertain changes
- Precipitation associated with ETCs will increase

Adaptation and Mitigation

Adaptation is to get prepared for the impacts to reduce the vulnerability to climate change (IPCC AR6 WG2):

- Vulnerability assessment
- Preparation for stronger floods and storms
- Agricultural and ecosystem adaptation
- Policy planning related to human health and air quality

Mitigation is to reduce anthropogenic CO₂ emissions and/or remove CO₂ from atmosphere (IPCC AR6 WG3):

- Climate engineering or geoengineering
- Renewable energy sources
- Reduce energy consumption and carbon footprint, and use energy efficiently